

PulseLM: A Foundation Dataset and Benchmark for PPG-Text Learning

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Abstract

Photoplethysmography (PPG) is a widely used non-invasive sensing modality for continuous cardiovascular and physiological monitoring across clinical, laboratory, and wearable settings. While existing PPG datasets support a broad range of downstream tasks, they typically provide supervision in the form of numerical measurements or task-specific labels, limiting their compatibility with language-based interfaces and multimodal foundation models. In this work, we introduce PulseLM, a large-scale PPG-text question-answering dataset that bridges raw PPG waveforms and natural language through a unified question-answering (QA) formulation. PulseLM aggregates PPG recordings from sixteen publicly available sources and harmonizes heterogeneous annotations into 12 downstream tasks. The dataset comprises over 1 million standardized 10-second PPG segments, associated with nearly 2.5 million question-answer pairs. We further define reproducible data pipeline, training, and evaluation protocols and establish baseline benchmarks using multimodal PPG-aware large language models. PulseLM provides a standardized foundation for studying language-grounded physiological inference, cross-dataset generalization, and scalable benchmarking of PPG-based multimodal models. We publicly release the dataset and code at Hugging Face and GitHub, respectively.

1 Introduction

Photoplethysmography (PPG) is a non-invasive optical sensing modality that underpins a wide range of modern health monitoring systems, from clinical pulse oximeters Lee et al. (2022) to consumer wearables such as smartwatches and earbuds app (2026); emp (2026); ear (2026). In principle, PPG measures changes in blood volume within the peripheral vasculature by emitting light onto the skin using a light-emitting diode (LED) and detecting variations in the intensity of transmitted or reflected light with a photodetector, where cardiac-induced pulsatile blood flow gives rise to a characteristic waveform. From that, PPG waveforms can be analyzed to enable continuous estimation of a broad range of cardiovascular and respiratory indicators, including heart rate (HR) Reiss et al. (2019), heart rate variability (HRV) Hu et al. (2025); Xu et al. (2025), respiratory rate (RR) Liu et al. (2012), blood pressure (BP) Li et al. (2024), stress levels Schmidt et al. (2018), and sleep-disordered breathing (SDB) Garde et al. (2014). Owing to its low cost, ease of deployment, and applications, PPG has become one of the most widely collected physiological signals worldwide.

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From a machine learning perspective, this abundance of PPG data presents a significant opportunity for developing physiological models that can operate across tasks, devices, and recording environments. In fact, recent advances in deep pretrained PPG foundation models (e.g., Papagei and PulsePPG) have demonstrated their effectiveness on diverse PPG downstream tasks Saha et al. (2025); Pillai et al. (2025); Abbaspourazad et al. (2024). In particular, these approaches leverage a pretrained single-modal PPG encoder to enable rapid and data-efficient development of downstream models. However, this paradigm still requires the post hoc development of task-specific models through linear probing or fine-tuning, while offering limited semantic interpretability and user interaction.

In the context of modern multimodal foundation models, such as in domains like natural language processing and computer vision, large language models (LLMs) assisted models have demonstrated that language can serve as a unifying interface for reasoning, explanation, and cross-task generalization SELLERGRÉN et al. (2025); LIU et al. (2023); PLAAT et al. (2025). Specifically for physiological sensing, similar advances have been enabled by large-scale electrocardiogram (ECG)-text and image-report datasets, which allow models to align raw signals with clinical semantics expressed in natural language LIU RUOQI and YUE XIANG (2026); OH et al. (2023); PHAM et al. (2025b); ZHANG et al. (2026), significantly improving the utility of these models. However, the development of such models requires signal-text-aligned datasets across different PPG downstream tasks, which are *absent in the context of PPG research*.

One promising way to bridge this gap is to formulate PPG tasks as a single LLM-based question-answering (QA) problem. Question answering provides a natural and flexible interface that mirrors how clinicians and end users interact with physiological data, for example, by asking whether a heart rate is normal, whether signal quality is sufficient, or whether a recording indicates elevated cardiovascular risk. Importantly, a QA formulation can unify heterogeneous downstream tasks under a single supervision paradigm, avoiding the need for task-specific regression heads while enabling models to associate continuous waveform patterns with discrete, interpretable physiological concepts.

However, enabling PPG-centric question answering at scale poses substantial challenges. Specifically, *public PPG datasets are highly fragmented, differing in sensor placement* (fingertip vs. wrist vs. ear), *sampling rate* (60 Hz vs. 128 Hz), *recording environment* (clinical vs. in-the-wild, stationary vs. motion), *subject population* (patients vs healthy, young vs. elderly people), and *available annotations* (heart rate, stress level, motion status, etc.). *Moreover, none of these datasets natively provides natural-language descriptions and QA supervision*. As a result, there is currently no large-scale, standardized benchmark that supports QA interfaces over PPG signals across diverse physiological domains and real-world conditions.

To address this issue, we introduce PulseLM, the first large-scale PPG-Text QA dataset and benchmark designed to support multimodal PPG learning and interpretation. PulseLM aggregates and harmonizes PPG recordings from diverse public sources spanning clinical, laboratory, and in-the-wild environments. For each standardized PPG segment, we derive physiologically grounded categorical labels from existing annotations or co-recorded reference signals (e.g., ECG) and convert them into QA supervision using controlled natural-language question-answer pools. The resulting dataset comprises over 1 million PPG-Text samples and nearly 2.5 million closed-ended question-answer pairs covering a broad range of rhythm, cardiovascular, stress, and respiratory tasks.

In addition to releasing the dataset, we establish a unified benchmarking protocol for PPG question answering and provide baseline results using multimodal models that integrate PPG encoders with instruction-tuned language models. By framing PPG interpretation as a language-based QA problem, PulseLM enables systematic evaluation of unified physiological inference across tasks, datasets, and recording conditions, and lays the foundation for future research on multimodal foundation models for PPG signals.

This work makes the following contributions:

- To the best of our knowledge, we introduce the first dataset to unify multiple heterogeneous PPG sources spanning clinical, laboratory, and in-the-wild environments into a single language-based QA benchmark. The dataset comprises over 1 million PPG samples and nearly 2.5 million question-answer pairs, covering 12 downstream tasks.
- We define corresponding benchmarks for multimodal PPG question-answering tasks, using consolidated PPG-language models that integrate pretrained PPG encoders with instruction-tuned LLMs to enable unified inference within a single QA framework.

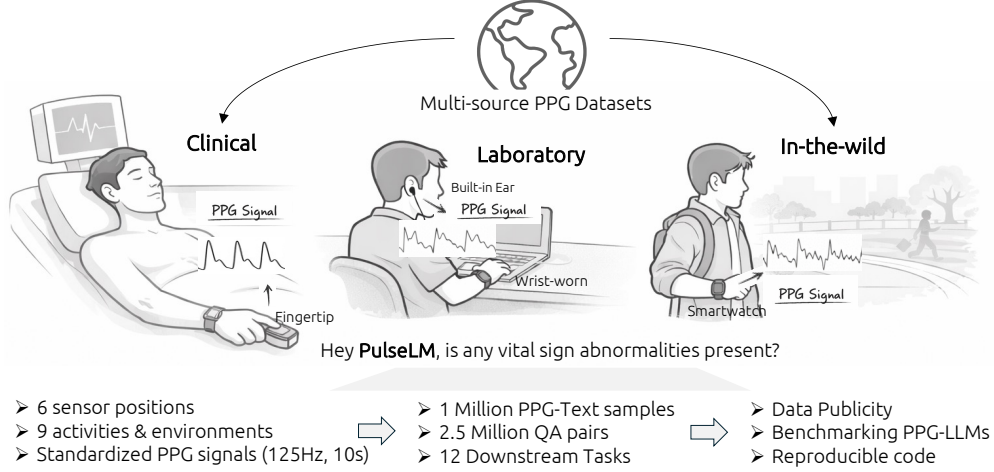


Figure 1: Overview of our dataset study.

- We release a unified data processing pipeline, model training, and evaluation protocols to facilitate future research on language-based PPG understanding.

2 Background

2.1 PPG Downstream Tasks

PPG measures volumetric changes in blood within peripheral tissue by illuminating the skin with a light source and recording variations in the reflected or transmitted light. Owing to this optical measurement principle, the resulting waveform is inherently sensitive to both physiological conditions and sensing-related factors. Under stable clinical conditions, such as fingertip recordings with minimal motion, PPG typically exhibits quasi-periodic cardiac cycles with well-defined morphological components, including a dominant systolic peak and a secondary diastolic peak associated with vascular reflections. However, in practical wearable scenarios, factors such as user movement Wang et al. (2019) or a loose contact Ho et al. (2025) between a wrist-worn device and the skin can disrupt sensor-skin coupling and deform underlying tissue, leading to severe waveform distortion, attenuation of pulsatile components, or even the disappearance of characteristic peaks.

However, with the development of signal processing techniques Peng et al. (2014); Deric Tang et al. (2016) and, more recently, deep learning methods Zargari et al. (2023); Kazemi et al. (2025); Reiss et al. (2019); Pillai et al. (2025); Pham et al. (2025a), PPG has enabled a broad spectrum of downstream tasks, including heart rate (HR), heart rate variability (HRV), and respiratory rate (RR) estimation, which are now routinely deployed in consumer wearables. These tasks rely primarily on temporal characteristics of the waveform and form the foundation of many real-world PPG-based systems. Beyond those core metrics, more complex inference tasks depend heavily on subtle waveform morphology. Blood pressure (BP) estimation Pan et al. (2024); González et al. (2023), including both systolic and diastolic pressure, has been extensively studied using PPG-derived features/representation, either through regression to continuous values or classification into clinically meaningful categories (e.g., hypertension stage 1).

In a related vein, rhythm-level cardiac inference, such as arrhythmia detection, exploits irregular inter-beat intervals, variable pulse amplitudes, and disrupted waveform morphology captured by PPG. These tasks require models to capture fine-grained morphological and rhythm structure and longer-range temporal dependencies within the signal. More recently, PPG has been explored in a range of non-traditional and higher-level inference domains. Studies have demonstrated its utility for emotion (stress) assessment Schmidt et al. (2018), sleep-disordered breathing (SDB) Garde et al. (2014) detection. Inference in these domains extends beyond purely signal-level pattern recognition and demands more expressive representations of physiological states.

Table 1: Consolidated statistics of the PulseLM dataset constructed from sixteen public PPG sources.

No.	Dataset	# PPG Segments	# QA Cats	# QA Pairs	QA Categories	Sensor Positions	Environment
(1)	VitalDB Lee et al. (2022)	163,959	5	819,441	HR, BP, HRV	finger	clinical
(2)	UCI Kachuee et al. (2015b)	111,751	3	335,253	HR, BP, SQI	finger	clinical
(3)	BCG Li et al. (2024)	671	3	2,013	HR, BP, SQI	finger	clinical
(4)	PPG-BP Liang et al. (2018)	369	2	738	HR, BP	finger	clinical
(5)	SDB Garde et al. (2014)	258,897	1	258,897	SDB	finger	clinical
(6)	Sensors Aguirre et al. (2021)	2,061	3	6,183	HR, BP, SQI	finger	clinical
(7)	UQVitalSigns Liu et al. (2012)	37,018	5	153,659	HR, BP, SpO ₂ , RR, SQI	finger	clinical
(8)	PPGArrhythmia Liu et al. (2022)	46,827	1	46,827	Arrhythmia	finger	clinical
(9)	MIMIC PERform Charlton et al. (2022)	4,196	1	4,196	AF (Binary)	finger	clinical
(10)	BIDMC Pimentel et al. (2016b)	12,462	2	24,924	SpO ₂ , RR	finger	clinical
(11)	EarSet Montanari et al. (2023)	1,776	1	1,776	HR	ear	lab
(12)	UTSA-PPG Xu et al. (2025)	16,925	4	67,700	HR, HRV	wrist, finger	lab
(13)	WESAD Schmidt et al. (2018)	2,998	1	2,998	Stress	wrist	lab
(14)	DALIA Reiss et al. (2019)	39,216	1	39,216	HR	wrist	in-the-wild
(15)	WildPPG Meier et al. (2024)	240,000	4	574,544	HR, HRV	wrist, mix	in-the-wild
(16)	AFPPGECG Bacevičius et al. (2024)	140,436	1	140,436	AF (Binary)	wrist	in-the-wild
Total	16 source datasets	1,079,562	12	2,478,801	mix	mix	mix

2.2 Existing PPG Datasets and Their Limitation

To date, a wide range of publicly available PPG datasets have been released over the past decades to support research in health monitoring and wearable sensing, spanning clinical, laboratory, and in-the-wild recording environments. For example, clinical datasets Lee et al. (2022); Kachuee et al. (2015a); Pimentel et al. (2016a) are the most common category, collected in hospitals or clinics daily for a large number of patients, and typically use fingertip pulse oximeters to acquire high-fidelity PPG signals with minimal motion interference. Laboratory datasets extend this setting by adopting wearable form factors, most commonly wrist-worn or built-in ear devices Schmidt et al. (2018); Montanari et al. (2023), and introducing structured protocols or controlled activities (e.g., mild physical movement such as sitting or talking) to study physiological responses under reproducible and specialized conditions. In contrast, in-the-wild or daily activity datasets Meier et al. (2024); Reiss et al. (2019) further improve ecological validity by capturing longer-term PPG recordings during unconstrained everyday activities (e.g., running, playing sports) using consumer-grade wearables, at the cost of increased motion artifacts and sensing variability.

Despite the diversity of available PPG datasets and their applications, existing resources exhibit several fundamental limitations when viewed through the lens of unified physiological understanding. Most datasets are designed around a limited downstream task and provide supervision in the form of numerical values or categorical labels specific to that task, such as heart rate values (e.g., 65 bpm) for heart rate estimation tasks. Consequently, models trained on these datasets are typically optimized for narrow objectives and lack a unified interface for cross-task inference.

A second limitation arises from dataset fragmentation. Public PPG datasets differ substantially in sensor placement, sampling rate, signal length, preprocessing pipelines, and label definitions. Even for ostensibly similar tasks, such as heart rate variability estimation, different datasets rely on distinct metrics, time windows, or reference standards. This heterogeneity makes it difficult to combine datasets or evaluate models under consistent protocols, and often leads to models that implicitly overfit to dataset-specific characteristics rather than learning generalizable physiological representations.

Finally, and most critically for language-enabled modeling, existing open-source PPG datasets lack natural-language supervision. Unlike ECG or medical imaging domains, where large-scale signal-text or image-report datasets Gow et al. (2023); Liu Ruoqi and Yue Xiang (2026) have enabled models to learn clinical semantics through language, PPG datasets remain confined to numeric or concretized categorical labels. *This absence of language-based supervision precludes the development of models that can describe waveform characteristics, explain physiological states, or interact with users through natural-language queries.* Taken together, these points suggest that progress in PPG-based learning is constrained not only by modeling approaches but also by the structure and scope of available datasets.

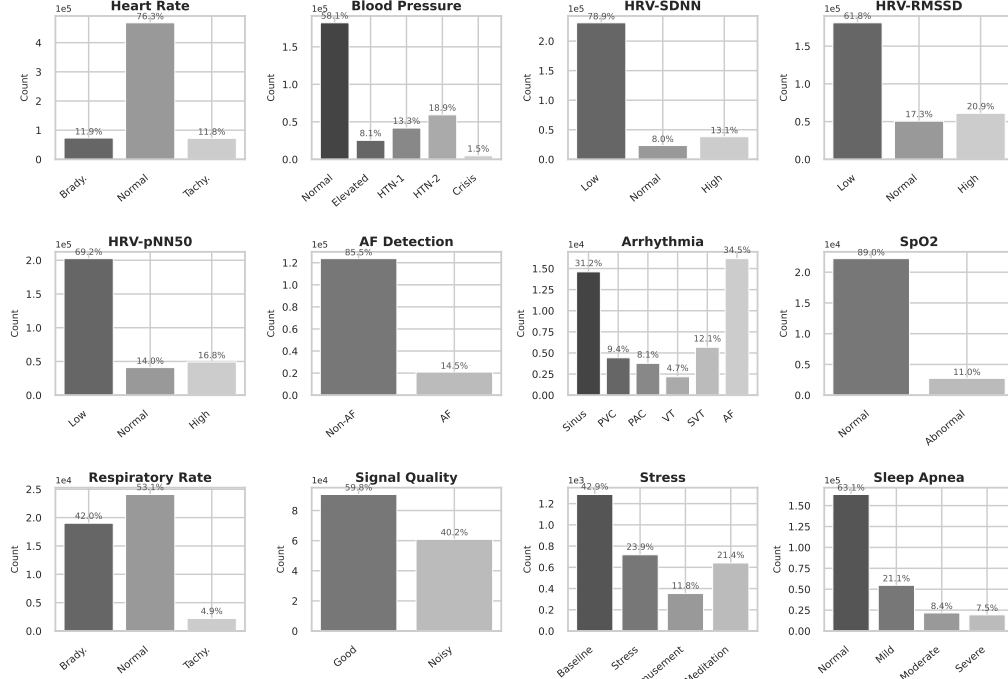


Figure 2: Demonstration of label distributions in PulseLM dataset.

3 PulseLM Dataset

3.1 Overview

We introduce PulseLM, a large-scale PPG-Text question-answering dataset designed to enable language-based physiological understanding across diverse tasks and recording conditions (see Figure 1). The dataset aggregates PPG recordings from 16 publicly available sources and reformulates their existing annotations into a unified QA representation. In total, PulseLM contains 1,079,562 standardized PPG segments paired with 2,478,801 physiologically grounded question-answer pairs spanning twelve unique QA categories. Each PPG segment is represented as a 10-second window and may be associated with multiple QA pairs depending on the availability of annotations in the source dataset. By expressing heterogeneous downstream tasks under a shared QA interface, PulseLM enables systematic benchmarking of multimodal models on unified physiological inference rather than task-specific prediction. To this end, the dataset spans clinical, laboratory, and in-the-wild wearable recordings, with PPG signals collected across diverse sensor placements such as finger pulse oximeters, wrist-worn wearables, in-ear sensors, and multi-site configurations. This diversity allows evaluation of robustness and generalization under realistic deployment scenarios. We will delve into key parts in the dataset below.

3.2 Source Dataset Selection

PulseLM leverages PPG recordings from publicly available datasets selected to maximize overall dataset diversity. A consolidated summary of dataset-level statistics, including data size, sensor placement, and environment, is provided in Table 1. Across all sources, the dataset supports 12 physiological QA categories spanning heart rate, blood pressure, heart rate variability, atrial fibrillation (binary), arrhythmia, SpO₂, respiratory rate, sleep-disordered breathing, stress, and signal quality assessment. It further covers finger, wrist, ear, and other multi-location PPG acquisition, as well as a wide range of device characteristics. Taken together, this diversity is a deliberate design choice to discourage overfitting to dataset-specific artifacts and to promote learning of transferable physiological representations.

3.3 Signal Standardization

To enable consistent multimodal learning across heterogeneous data sources, all PPG recordings are standardized through a unified preprocessing pipeline comprising four stages: (1) *Resampling*, where signals are resampled to a common rate of 125 Hz to eliminate variability across devices and protocols; (2) *Filtering*, where a fourth-order Butterworth low-pass filter (8 Hz cutoff) is applied to suppress high-frequency noise, followed by DC (non-pulsatile) removal via per-segment mean subtraction to reduce baseline drift while preserving the hemodynamically relevant AC (pulsatile) component; (3) *Segmentation*, where signals are partitioned into fixed-length 10-second windows, which are widely used to support PPG downstream tasks Pillai et al. (2025); Saha et al. (2025); and (4) *Normalization*, where each segment is independently scaled to the $[0, 1]$ range using min-max normalization to mitigate amplitude variations while preserving waveform morphology and temporal structure.

3.4 Ground Truth Harmonization

In our study, PulseLM derives labels from original dataset annotations or from reference signals co-recorded with PPG, such as ECG. Specifically, depending on the source, available ground truth includes continuous-valued physiological measurements (e.g., blood pressure, SDB, and stress), categorical clinical labels (e.g., atrial fibrillation or arrhythmia type), contextual states (e.g., activity or stress), and signal quality indicators. For datasets that provide synchronized ECG recordings (e.g., VitalDB, WildPPG, and UTSA-PPG), HR and HRV labels (including ultra-short RMSSD: Root Mean Square of Successive Differences, SDNN: Standard Deviation of Normal-Normal intervals, and pNN50: Percentage of NN intervals differing by more than 50 ms) are computed using established ECG-based signal processing toolkits Makowski et al. (2021), treating ECG as the gold standard for rhythm and variability estimation. Ultimately, we provide the label distributions across tasks in Figure 2.

Next, to enable unified language-based supervision across heterogeneous datasets, all ground truth is harmonized into a shared categorical label space using predefined mapping rules. Continuous-valued measurements are discretized into clinically or physiologically meaningful categories (e.g., bradycardia/normal/tachycardia for heart rate, standard risk-hypertension stages for blood pressure), including ECG-derived HR and HRV metrics where applicable. Dataset-provided categorical labels are preserved when compatible and otherwise normalized to a consistent label set. All mapping rules and thresholds are fixed *a priori* and applied deterministically; the complete specifications are provided in Appendix A.2. Through this process, PulseLM supports twelve QA tasks spanning cardiovascular state, rhythm analysis, variability assessment, signal quality, behavior and affect, respiratory function, and sleep-related risk. All QA supervision is grounded in original annotations or reference signals, and language models infer no labels during dataset construction.

3.5 Question Answering Formulation

PulseLM represents physiological supervision using a QA formulation, in which each PPG segment is paired with natural-language questions and answers. Depending on the availability of annotations in the source dataset, a single PPG segment may be associated with multiple QA instances spanning different physiological domains. For each QA category, a predefined answer vocabulary is specified based on the harmonized physiological labels described in Section 3.4. This formulation constrains model outputs to a finite and interpretable set of responses, enabling consistent supervision and objective evaluation across heterogeneous datasets.

To introduce linguistic variability while preserving semantic consistency, questions are generated for each QA category using controlled templates and paraphrasing. Specifically, we leverage OpenAI’s GPT-5 with a strict instruction prompt to output candidate rephrasings, which are subsequently manually verified and all covered in Table 5, to ensure semantic equivalence and quality. Furthermore, all answers are deterministically assigned by mapping existing ground-truth annotations to the corresponding categorical labels. As a result, the QA construction procedure does not introduce new or hallucinatory physiological information. Following that, to facilitate clear, reproducible benchmarking, we partition the data into train, validation, and test splits using subject-disjoint separation applied per source dataset, ensuring no subject overlap across splits while keeping the split

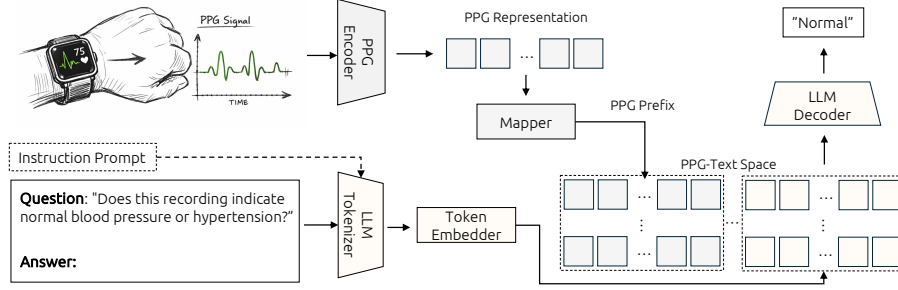


Figure 3: Illustration of benchmarking PPG-LLM baselines.

ratio generally around 8/1/1. Importantly, paraphrases in the test split are strictly disjoint from those used during training. Full data examples and split statistics are provided in Appendix A.

4 PulseLM Benchmarking

4.1 Objectives and Scope

Having established the PulseLM dataset construction and QA formulation in Section 3, we now turn to the central empirical question: *can multimodal models learn to ground natural-language physiological questions in raw PPG signals?* To this end, our goal is to provide an initial evaluation of language model behavior across diverse PPG dataset domains (e.g., sensor positions or recording environments) captured by PulseLM.

Accordingly, our benchmarking pursues two complementary objectives. First, we establish reproducible in-domain baselines by training and evaluating models on matched subject-wise train-test splits from the same source datasets, providing a direct measure of how well current multimodal architectures can exploit the unified QA supervision that PulseLM provides. Second, we employ cross-dataset generalization by training on a single source dataset and evaluating on entirely held-out sources with different sensor placements and recording environments. This setting directly tests whether models learn transferable physiological representations or merely overfit to dataset-specific signal characteristics. Exact-match (EM) accuracy is used as the strict metric throughout for the text generation task, supplemented by per-dataset breakdowns to reveal where models succeed and where they fall short. All benchmarks operate over the harmonized label space described in Section 3, ensuring consistent evaluation across heterogeneous sources.

4.2 Baseline Models

To establish a comprehensive reference for performance on PulseLM, we benchmark two complementary families of models: non-LLM transformer baselines that isolate the contribution of individual modalities, and full multimodal PPG-language models that form the primary focus of this benchmark.

Non-LLM Baselines. Before attributing performance to multimodal grounding, it is essential to quantify what each modality contributes independently as natural baselines for our tasks. Inspired by the experimental design in ECG-QA Oh et al. (2023), we include non-LLM baselines that systematically vary modality access. First, the Blind model processes only the natural-language question through a frozen BERT encoder Devlin et al. (2019) and a linear classification head, without any access to the PPG signal. The Deaf model takes the complementary perspective: it operates exclusively on the PPG signal via a pretrained PPG encoder and a linear classifier, receiving no textual input. Lastly, the Fusion model integrates both modalities without an LLM: a PPG encoder and a BERT text encoder produce independent representations, which are concatenated and passed through a multi-layer perceptron to predict the answer. All baselines predict over a shared global answer vocabulary across all 12 QA categories. Together, they constitute an interpretable modality-ablation suite, enabling us to determine whether LLM-based models exploit physiological signals or rely on language shortcuts.

Table 2: In-domain evaluation using EM accuracy across different datasets. Columns correspond to individual datasets (1)-(16) following the order in Table 1.

Language Models	PPG Encoder	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	Avg. $\pm$ Std
Blind PPG	None	52.6	54.8	58.3	62.8	24.1	66.0	61.0	28.9	48.8	55.4	88.5	52.1	18.3	83.7	65.9	86.1	56.7 $\pm$ 19.8
Deaf PPG	Papagei	49.0	42.4	17.2	42.3	25.8	45.3	52.8	13.0	50.8	87.9	79.9	44.5	0.0	61.8	35.0	81.2	45.5 $\pm$ 23.8
	PulsePPG	46.8	44.8	40.1	66.7	51.0	44.5	47.8	42.6	9.6	61.3	45.6	25.8	1.7	51.0	48.3	53.3	42.6 $\pm$ 16.5
Fusion PPG	Papagei	71.9	60.5	73.4	60.3	36.4	67.6	64.3	28.5	45.2	84.2	72.3	53.1	23.3	65.9	70.0	86.1	60.2 $\pm$ 17.9
	PulsePPG	55.8	71.2	74.0	47.4	43.0	73.3	71.3	65.6	44.4	69.9	75.3	59.1	41.5	68.5	76.6	85.7	63.9 $\pm$ 13.2
Qwen3-4B	Papagei	77.1	52.9	74.0	69.2	61.0	48.0	66.7	48.5	49.9	88.2	88.5	63.9	39.5	83.7	85.3	86.1	67.7 $\pm$ 15.9
	PulsePPG	74.8	68.5	32.3	69.2	54.9	72.4	75.9	52.9	65.8	87.9	89.6	65.7	44.4	84.5	85.4	85.2	69.3 $\pm$ 16.0
LLaMA3.2-3B-Instruct	Papagei	77.6	47.6	74.0	69.2	61.0	42.9	75.0	48.4	49.9	88.2	88.5	63.9	39.5	83.7	85.3	86.1	67.6 $\pm$ 16.8
	PulsePPG	75.0	69.0	38.5	69.2	55.9	71.2	74.9	51.4	71.8	86.6	90.1	65.9	43.7	85.1	85.2	84.9	69.9 $\pm$ 15.1
LLaMA3.1-8B-Instruct	Papagei	77.9	53.6	74.0	69.2	61.0	54.3	75.0	42.7	49.9	88.2	88.5	63.9	39.5	83.7	85.3	86.1	68.3 $\pm$ 16.0
	PulsePPG	74.7	69.4	49.5	69.2	58.4	70.8	74.3	49.6	75.0	81.3	89.6	65.7	47.4	85.0	85.2	84.5	70.6 $\pm$ 13.1
Qwen2.5-7B-Instruct	Papagei	77.5	55.5	64.6	69.2	61.0	49.5	70.9	48.4	49.9	68.5	88.5	63.9	39.5	83.7	85.3	86.1	66.4 $\pm$ 14.7
	PulsePPG	75.4	77.8	79.2	69.2	54.1	75.1	72.9	59.6	62.9	75.6	90.7	65.1	49.7	86.2	85.1	85.9	72.8 $\pm$ 11.6

Multimodal PPG-Language Models. In addition to those baselines, we adopt a unified architecture, illustrated in Figure 3, that couples a pretrained PPG encoder with an instruction-tuned LLM via a lightweight projection layer, enabling joint modeling of physiological signals and textual queries without modifying the LLM structure. Specifically, a PPG encoder first processes the 10-second input waveform, producing a 512-dim latent embedding that encapsulates physiological features. A linear projection layer then maps this embedding into the LLM’s token embedding space, yielding a PPG prefix token that is prepended to the embedded question tokens. The resulting multimodal sequence is fed into the LLM decoder, which attends jointly over both modalities to generate the answer.

A central design choice in this architecture is the PPG encoder, as it determines the quality of physiological features passed to the language model. This choice is particularly important given the heterogeneity of PPG signals across datasets (e.g., sensor placement, motion artifacts, and acquisition conditions), which imposes strong demands on the encoder’s ability to learn robust, transferable representations. Therefore, we employ two well-established pretrained PPG encoders: Papagei Pillai et al. (2025), trained on large-scale PPG data at 125 Hz, and PulsePPG Saha et al. (2025), pretrained at 50 Hz (signals are resampled accordingly). Pairing each LLM backbone with both common encoders allows us to examine the effect of physiological representation quality from that of language model capacity.

For the LLM backbone, we evaluate a set of models: Qwen3-4B, Qwen2.5-7B-Instruct, LLaMA3.2-3B-Instruct, and LLaMA3.1-8B-Instruct. This selection balances model capacity with computational feasibility and provides an initial investigation into how model scale and backbone design affect physiological grounding performance using the proposal PulseLM dataset.

4.3 Implementation Details

To ensure fair and reproducible comparison across all model variants, we adopt a unified training protocol for all models. Because the pretrained PPG encoders already capture rich physiological representations, we freeze their parameters throughout fine-tuning, adapting only the projection layer and the language model. For language model adaptation, we employ low-rank adaptation (LoRA) Hu et al. (2022) with rank $r = 8$ and scaling factor $\alpha = 16$ (dropout 0.1). The projection layer is trained with a slightly higher learning rate of 2×10^{-4} to encourage faster alignment, while LoRA parameters use 1×10^{-4} . Regarding the instruction prompt, we present it in detail from Box B. All models are optimized with AdamW using a cosine learning rate schedule with a linear warm-up over the first 3% of training steps, and trained for 2 epochs with a batch size of 16 and 4-step gradient accumulation in bfloat16 mixed precision on a single NVIDIA H200 GPU (140GB VRAM). For evaluation, all models use greedy decoding with a maximum of 32 new tokens. Predicted answers are extracted by parsing the <answer> tag in the generated output; if no valid tag is found or the text does not match any option, the prediction is treated as incorrect. All hyperparameters, training scripts, and evaluation code are released alongside the dataset.

Table 3: Cross-dataset evaluation: models trained on source datasets and evaluated on held-out datasets. The PPG encoder used is PulsePPG (except blind), and results are reported with 95% bootstrap confidence intervals.

Model	VitalDB			WildPPG		
	BCG	UTSA-PPG	EarSet	DALIA	UTSA-PPG	EarSet
Blind PPG	60.9 $\pm$ 8.5	44.4 $\pm$ 0.9	88.5 $\pm$ 3.3	65.9 $\pm$ 1.3	59.1 $\pm$ 0.9	72.3 $\pm$ 4.6
Deaf PPG	43.0 $\pm$ 8.6	34.0 $\pm$ 0.9	47.0 $\pm$ 5.1	68.6 $\pm$ 1.2	46.6 $\pm$ 0.9	79.7 $\pm$ 4.1
Fusion PPG	60.9 $\pm$ 8.5	44.4 $\pm$ 0.9	86.0 $\pm$ 3.6	68.7 $\pm$ 1.2	59.4 $\pm$ 0.9	73.4 $\pm$ 4.5
Qwen2.5-7B-Instruct	60.9 $\pm$ 8.5	49.5 $\pm$ 0.9	89.0 $\pm$ 3.2	85.3 $\pm$ 1.0	63.9 $\pm$ 0.9	88.5 $\pm$ 3.3

4.4 Main Results

In-domain evaluation. Table 2 reports results across all 16 datasets. First, a clear performance gap can be observed between non-LLM baselines and multimodal PPG-language models. The non-LLM approaches, including Blind, Deaf, and Fusion PPG models, consistently achieve lower performance, with average accuracy generally below 63%. This suggests that relying on either language or signal alone, or combining them through simple feature fusion, is insufficient for capturing the full range of physiological semantics in PulseLM. In contrast, multimodal PPG-language models deliver generally stronger results across datasets. On average, these models improve performance by approximately 6%-9%, reaching around 70% accuracy. In particular, Qwen2.5-7B-Instruct paired with PulsePPG achieves over 72% average accuracy, representing a substantial gain over all non-LLM baselines.

We also observe that across multimodal LLM architectures with different PPG encoders, PulsePPG generally yields better performance than Papagei, and both outperform non-LLM baselines. This suggests two points. First, integrating pretrained PPG encoders with language models is an effective design choice, as both encoders lead to meaningful performance gains within this framework. Second, the advantage of PulsePPG may stem from its pretraining on more diverse datasets spanning both laboratory and in-the-wild settings, including various levels of motion-affected samples, whereas Papagei is primarily trained on more controlled clinical data, which may limit its robustness under heterogeneous conditions. Another important factor in our framework is the choice of LLM backbone. Although all LLM-based models outperform non-LLM baselines, performance varies across architectures. Models based on the Qwen family tend to achieve slightly better overall results compared to LLaMA, indicating that backbone design and pretraining strategy also influence multimodal physiological inference performance.

Cross-dataset evaluation. In this section, we present additional analysis of cross-dataset generalization, focusing on model transfer across datasets with varying populations, devices, and acquisition conditions. Table 3 evaluates two representative cross-domain scenarios using VitalDB and WildPPG as training datasets, and compares our best-performing model (i.e., Qwen2.5-7B-Instruct with PulsePPG) against predefined non-LLM baselines. In the VitalDB scenario, models are trained on the clinical finger-sensor data and evaluated on wearable datasets with distinct sensor placements or environments (EarSet, BCG, UTSA-PPG). This setting investigates robustness to cross-dataset variation when transferring from controlled clinical data to more diverse conditions. Similarly, in the WildPPG scenario, models are trained on large-scale in-the-wild wrist recordings and evaluated on other recording conditions (DALIA, UTSA-PPG, EarSet). This setting examines transfer from noisy, naturalistic data to more controlled environments.

Across both scenarios, we observe trends generally consistent with the in-domain evaluation, with the multimodal LLM-based approach consistently outperforming the non-LLM baselines. Despite substantial differences in sensor placement and acquisition conditions, the models maintain relatively competitive performance across several held-out datasets, suggesting that the learned representations capture transferable physiological information beyond dataset-specific and participant-specific patterns. At the same time, the variability across transfer settings indicates that cross-domain generalization in PPG remains sensitive to dataset heterogeneity. Overall, these findings reinforce the role of PulseLM as a benchmark for systematically studying representation transfer and robustness across diverse physiological sensing environments.

5 Conclusion

This paper presents PulseLM, a large-scale dataset designed to support multimodal physiological inference from PPG signals using language-based supervision. Consolidating heterogeneous PPG recordings from multiple sources into a unified closed-ended question-answering task, PulseLM enables consistent modeling and evaluation across diverse downstream inferences. The accompanying baseline results provide an initial reference, highlighting both the promise and current challenges of multimodal physiological LLMs under realistic data heterogeneity, task diversity, and domain shifts. We hope PulseLM to serve as a reference point for future work on scalable physiological foundation models over wearable biosignals.

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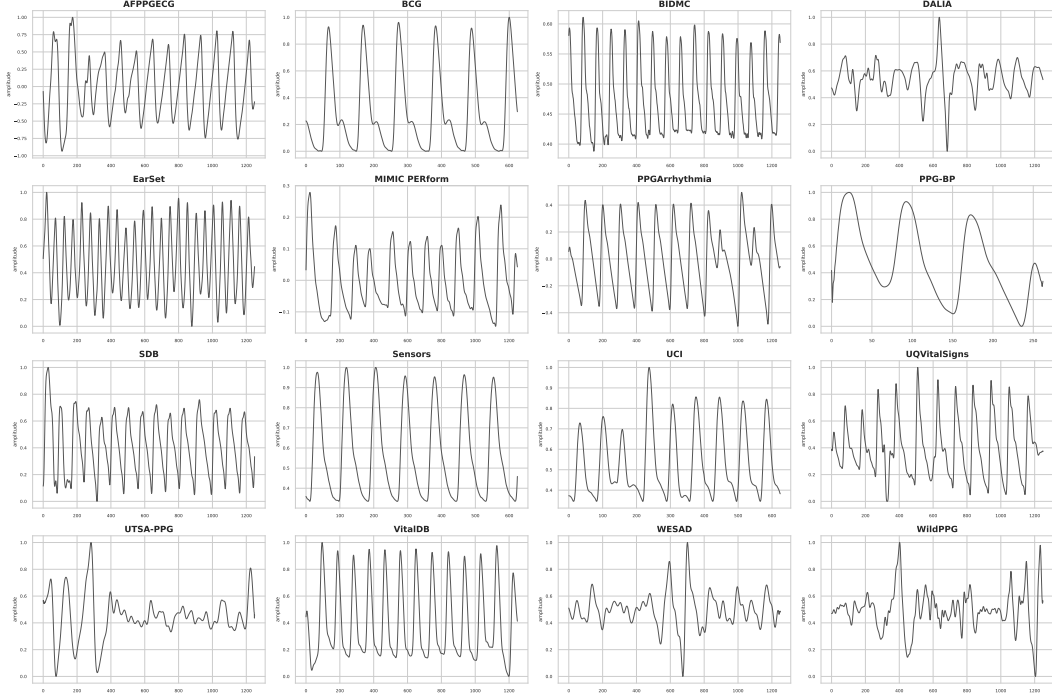


Figure 4: Illustrative PPG waveforms from the PulseLM train dataset. Each recording is windowed to 10 s; shorter signals are padded to match this fixed duration.

A Additional Dataset Details

This appendix provides supplementary material to support further dataset and benchmark details for reproducibility and transparency. Specifically, we include illustrative PPG waveform samples (Appendix A.1), the label harmonization rules (Appendix A.2), the full question templates per category (Appendix A.3), the dataset structure (Appendix A.4), and dataset split statistics (Appendix A.5). We present each of them in the sections below.

A.1 PPG Waveform Examples

First, Figures 4 and 5 show representative 10-second PPG waveforms from the train and test partitions of PulseLM. The examples span multiple source datasets and sensor placements, illustrating the considerable morphological diversity in the benchmark. Clinical finger-sensor recordings (e.g., VitalDB, BCG) tend to exhibit well-defined systolic peaks, while wrist-worn and in-ear recordings (e.g., WildPPG, Dalia) show broader, more attenuated morphologies. Signals captured during unconstrained activity exhibit visible motion artifacts and baseline drift, highlighting the challenge of cross-environment generalization.

A.2 Label Mapping

In our study, as the included PPG sources provide heterogeneous annotations (e.g., continuous numerical values such as blood pressure), we harmonize all physiological measurements into a unified set of categorical labels. The mapping rules are summarized in Table 4, which was derived from established clinical guidelines (e.g. AHA, AASM) where available and complemented with commonly used practices and literature to enable consistent large-scale benchmarking.

A.3 Question Templates

Importantly, we report the full set of question paraphrases for each QA category in Table 5, alongside the prompts used for generation (Box A.3) with GPT-5. Each category uses 10 templates for

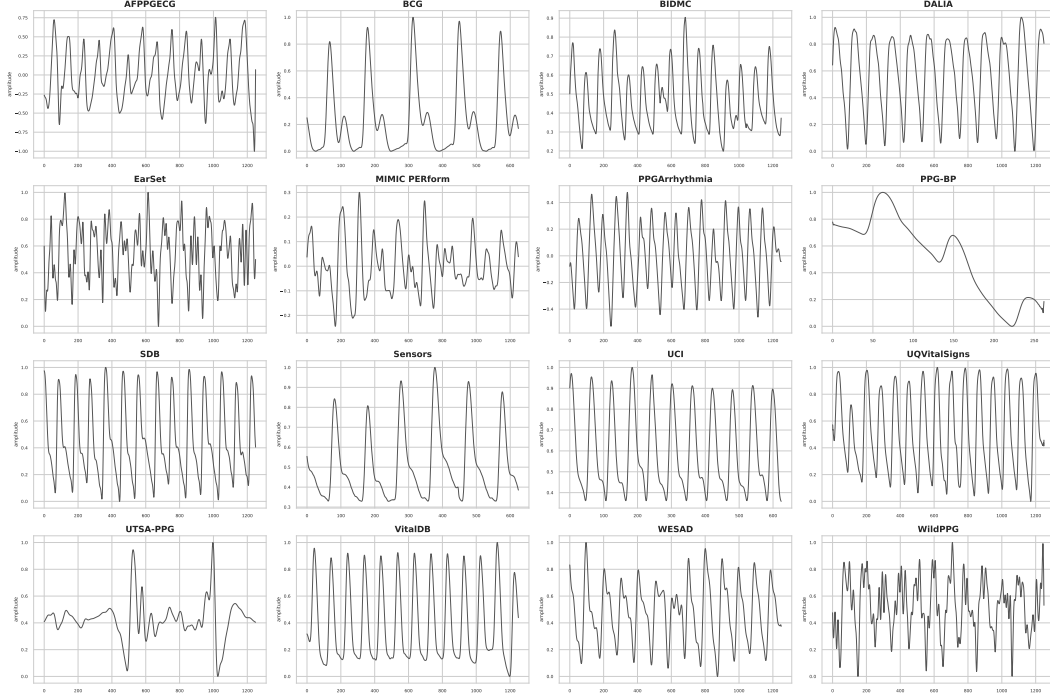


Figure 5: Illustrative PPG waveforms from the PulseLM test dataset. Each recording is windowed to 10 s; shorter signals are padded to match this fixed duration.

Table 4: Label mapping rules for physiological signal harmonization.

Category	Mapping rule
Heart rate	HR (bpm) is discretized into three classes: bradycardia (< 60), normal ($60\text{--}100$), and tachycardia (> 100).
Blood pressure	SBP/DBP (mmHg) are mapped using severity-prioritized rules: hypertensive crisis ($\text{SBP} > 180$ or $\text{DBP} > 120$), stage 2 ($\text{SBP} \geq 140$ or $\text{DBP} \geq 90$), stage 1 ($\text{SBP} \geq 130$ or $\text{DBP} \geq 80$), elevated ($\text{SBP} 120\text{--}129$ and $\text{DBP} < 80$), and normal ($\text{SBP} < 120$ and $\text{DBP} < 80$).
Signal quality	Skewness-based rule: $0.5 \leq \text{skew} \leq 2.0$ is good_quality ; otherwise noisy .
Sleep apnea	AHI (events/hour) is mapped as normal (< 5), mild ($5\text{--}14$), moderate ($15\text{--}29$), and severe (≥ 30).
HRV	RMSSD (ms): low (< 20), normal ($20\text{--}50$), high (> 50). SDNN (ms): low (< 50), normal ($50\text{--}100$), high (> 100). pNN50 (%): low (< 3), normal ($3\text{--}25$), high (> 25).
SpO₂	SpO ₂ (%) is categorized as normal (≥ 95), abnormal .
Respiratory rate	RR (breaths/min) is classified as bradypnea (< 12), normal ($12\text{--}20$), and tachypnea (> 20).

training/validation and 5 strictly disjoint paraphrases for testing, ensuring evaluation measures linguistic generalization rather than template memorization.

LLM Prompt for Question Template Generation per QA Category.

You are helping to build a PPG question-answering benchmark dataset. Based on the QA category and option below, generate 15 distinct paraphrased question templates.

Input. Category: \$\$ Options: \$\$

Output.

Table 5: Questions per category used during training/validation and test.

Category	Train & Validation Questions	Test Questions
AF Detection	1. Assess whether atrial fibrillation is present in this PPG. 2. Classify this PPG as AF or non-AF. 3. Determine whether this signal indicates atrial fibrillation. 4. Does this PPG signal show atrial fibrillation? 5. Does this waveform indicate AF? 6. Is atrial fibrillation present in this recording? 7. Is this a normal rhythm or atrial fibrillation? 8. Provide the atrial fibrillation detection result. 9. What is the AF detection result for this segment? 10. What is the AF label for this PPG recording?	1. Based on this PPG, report the rhythm status regarding AF. 2. Give the atrial fibrillation screening outcome for this window. 3. How would you characterize the cardiac rhythm in this recording? 4. Select the rhythm label: AF or sinus rhythm. 5. State the AF classification result for this signal.
Arrhythmia	1. Assess the arrhythmia classification from this recording. 2. Categorize the heart rhythm abnormality. 3. Classify the cardiac rhythm in this recording. 4. Determine the rhythm classification for this waveform. 5. Identify the arrhythmia type from the signal. 6. Is this a normal rhythm or arrhythmia? 7. What cardiac rhythm category does this sample belong to? 8. What is the arrhythmia category for this segment? 9. What is the rhythm diagnosis for this segment? 10. What type of arrhythmia does this signal show?	1. Give the arrhythmia assessment for this PPG window. 2. How would you characterize the rhythm disorder in this recording? 3. Name the cardiac rhythm pattern shown in this signal. 4. Select the most fitting rhythm label for this segment. 5. State the rhythm diagnosis based on this waveform.
Blood Pressure	1. Based on the PPG, what is the BP category? 2. Categorize the blood pressure level. 3. Classify the blood pressure level shown in this PPG segment. 4. Determine the BP classification from this waveform. 5. Does this sample indicate normal blood pressure or hypertension? 6. Provide the blood pressure risk category. 7. What blood pressure class does this sample belong to? 8. What hypertension stage does this PPG correspond to? 9. What is the blood pressure category for this sample? 10. What is the blood pressure status for this recording?	1. Give the blood pressure assessment for this recording. 2. How would you characterize the blood pressure in this signal? 3. Name the hypertension stage indicated by this PPG. 4. Select the BP classification that applies to this sample. 5. State the blood pressure diagnosis for this waveform.

Continued on next page

Table 5 – continued from previous page

Category	Train & Validation Questions	Test Questions
Heart Rate	1. Based on the PPG waveform, what is the HR category? 2. Categorize the heart rate shown in this recording. 3. Classify the heart rate based on this waveform. 4. Determine the heart rate category from the signal. 5. Is the heart rate normal, bradycardic, or tachycardic? 6. Provide the clinical heart rate category. 7. What heart rate classification does this PPG indicate? 8. What is the heart rate category for this PPG segment? 9. What is the heart rate status for this sample? 10. Which heart rate class does this sample belong to?	1. Give the heart rate assessment for this recording. 2. How would you characterize the heart rate in this signal? 3. Name the heart rate condition shown in this PPG. 4. Select the HR classification that applies to this sample. 5. State the heart rate diagnosis for this waveform.
HRV-pNN50	1. Assess the pNN50 category from this PPG. 2. Categorize the pNN50 variability measure. 3. Classify the pNN50 level. 4. Determine the pNN50 level for this recording. 5. How would you categorize pNN50 for this PPG? 6. Is pNN50 low, normal, or high in this sample? 7. Provide the pNN50 category. 8. What is the pNN50 category for this segment? 9. What is the pNN50-based HRV classification? 10. What pNN50 class does this sample belong to?	1. Give the pNN50 assessment for this recording. 2. How would you characterize the pNN50 in this signal? 3. Name the pNN50 condition shown in this PPG. 4. Select the pNN50 classification that applies to this sample. 5. State the pNN50 variability level for this waveform.
HRV-RMSSD	1. Assess the RMSSD-based variability level. 2. Categorize the short-term HRV (RMSSD). 3. Classify the RMSSD-based heart rate variability. 4. Determine the RMSSD classification. 5. How would you categorize RMSSD here? 6. Is the RMSSD low, normal, or high? 7. Provide the RMSSD category for this sample. 8. What RMSSD class does this recording indicate? 9. What is the HRV RMSSD category for this segment? 10. What is the parasympathetic activity level (RMSSD)?	1. Give the RMSSD assessment for this recording. 2. How would you characterize the RMSSD in this signal? 3. Name the RMSSD condition shown in this PPG. 4. Select the RMSSD classification that applies to this sample. 5. State the short-term HRV (RMSSD) level for this waveform.
HRV-SDNN	1. Assess the SDNN-based variability category. 2. Categorize the overall HRV (SDNN) level. 3. Classify the SDNN-based heart rate variability level. 4. Determine the SDNN level for this recording. 5. How would you categorize SDNN for this PPG? 6. Is the SDNN low, normal, or high in this sample? 7. Provide the SDNN category based on this PPG segment. 8. What SDNN class does this sample belong to? 9. What is the HRV SDNN category for this segment? 10. What is the SDNN-based HRV classification?	1. Give the SDNN assessment for this recording. 2. How would you characterize the SDNN in this signal? 3. Name the SDNN condition shown in this PPG. 4. Select the SDNN classification that applies to this sample. 5. State the overall HRV (SDNN) level for this waveform.
Respiratory Rate	1. Assess the respiratory rate category from this PPG. 2. Categorize the breathing rate. 3. Classify the respiratory rate level. 4. Determine the respiratory rate category for this recording. 5. How would you categorize the respiratory rate here? 6. Is the respiratory rate normal, slow, or fast? 7. Provide the respiratory rate category. 8. What is the breathing rate classification for this segment? 9. What is the respiratory rate category for this sample? 10. What respiratory rate class does this sample belong to?	1. Give the respiratory rate assessment for this recording. 2. How would you characterize the breathing rate in this signal? 3. Name the respiratory condition shown in this PPG. 4. Select the RR classification that applies to this sample. 5. State the respiratory rate diagnosis for this waveform.

Continued on next page

Table 5 – continued from previous page

Category	Train & Validation Questions	Test Questions
Sleep Apnea	1. Assess the respiratory disturbance category. 2. Categorize the breathing disorder level. 3. Classify the sleep breathing pattern. 4. Determine the sleep apnea severity. 5. Does this segment indicate sleep apnea? 6. Provide the SDB classification for this PPG window. 7. What is the AHI-based severity category? 8. What is the breathing disorder category? 9. What is the sleep-disordered breathing label for this segment? 10. What sleep-disordered breathing class is this?	1. Give the sleep-disordered breathing assessment for this recording. 2. How would you characterize the breathing disorder in this signal? 3. Name the sleep apnea severity shown in this PPG. 4. Select the SDB classification that applies to this sample. 5. State the AHI-based severity for this waveform.
SpO2	1. Assess the oxygen saturation level from this PPG. 2. Categorize the SpO2 level. 3. Classify the blood oxygen saturation. 4. Determine the SpO2 category for this recording. 5. How would you categorize SpO2 for this PPG? 6. Is the SpO2 normal or does it indicate hypoxemia? 7. Provide the oxygen saturation category. 8. What SpO2 class does this sample belong to? 9. What is the SpO2 category for this segment? 10. What is the oxygen saturation classification?	1. Give the SpO2 assessment for this recording. 2. How would you characterize the oxygen saturation in this signal? 3. Name the hypoxemia condition shown in this PPG. 4. Select the SpO2 classification that applies to this sample. 5. State the blood oxygen saturation diagnosis for this waveform.
Signal Quality	1. Assess the signal quality of this PPG recording. 2. Classify the PPG signal quality based on skewness. 3. Determine the signal quality index category. 4. How would you categorize the signal quality here? 5. Is this PPG recording of good or poor quality? 6. Is this PPG signal clean or motion distorted? 7. Provide the SQI quality category for this sample. 8. Rate the quality of this PPG signal. 9. What is the SQI classification for this segment? 10. What is the signal quality category for this PPG waveform?	1. Give the signal quality assessment for this recording. 2. How would you characterize the PPG signal integrity here? 3. Name the signal quality condition shown in this waveform. 4. Select the SQI classification that applies to this sample. 5. State the signal quality diagnosis for this PPG.
Stress	1. Categorize the stress condition for this sample. 2. Classify the stress level from this PPG. 3. Determine the emotional/stress state. 4. Identify the stress level for this segment. 5. Provide the stress state for this PPG window. 6. What is the affective state for this recording? 7. What is the emotional state label? 8. What is the stress label for this segment? 9. What psychological state does this segment indicate? 10. What stress category does this sample belong to?	1. Give the stress assessment for this recording. 2. How would you characterize the psychological state in this signal? 3. Name the stress condition shown in this PPG. 4. Select the stress classification that applies to this sample. 5. State the affective state diagnosis for this waveform.

A.4 Dataset Structure and QA Format

Each sample in PulseLM is stored as a row in a Parquet file and contains three key fields: 1) **signal**: a fixed-length array of 1,250 floats representing a 10-second PPG segment sampled at 125 Hz and normalized to $[0, 1]$; 2) **qa**: a JSON object mapping each applicable question category to a {question, answer} pair; and 3) **text**, an optional natural-language description of the recording context, which is derived from original dataset metadata, such as physiological labels, demographic attributes, activity context, sensor placement, and recording environment (Table 10). The qa field may contain multiple entries (i.e., 1-5) per sample, depending on which task labels are available for the recording. Box A.4 illustrates the schema of a PulseLM sample in our Hugging Face repository, showing how a single PPG segment is associated with multiple QA pairs across distinct physiological categories.

Sample schema in PulseLM.

```

signal:  $[x_1, x_2, \dots, x_{1250}]$ 
text: " $\langle \dots \rangle$ "
qa:
{
  "<category1>": { "question": "<...>", "answer": "<...>" },
  "<category2>": { "question": "<...>", "answer": "<...>" },
  "<category3>": { "question": "<...>", "answer": "<...>" },
  :
}

```

A.5 Dataset Split Statistics

To provide a more comprehensive view of our dataset, Tables 6-9 report detailed split statistics across four dimensions: PPG segment counts per source dataset (Table 6), QA pair counts by question category (Table 7), QA pair counts per source dataset (Table 8), and segment counts by recording environment (Table 9). VitalDB and WildPPG are the two largest contributors. We can also observe that clinical recordings represent 59% of the total data, and heart rate has the highest QA count while stress is the scarcest, reflecting the natural prevalence of these physiological states in the source datasets.

Table 6: Number of PPG signals per dataset and split.

Dataset	Train	Val	Test	Total
VitalDB	131,324	16,485	16,150	163,959
UCI	89,054	11,286	11,411	111,751
BCG	521	86	64	671
PPG-BP	294	36	39	369
SDB	205,618	23,525	29,754	258,897
Sensors	1,631	180	250	2,061
UQVitalSigns	25,202	2,703	9,113	37,018
PPGArrhythmia	36,820	4,764	5,243	46,827
MIMIC PERform	3,239	240	717	4,196
BIDMC	9,412	1,652	1,398	12,462
EarSet	1,368	44	364	1,776
UTSA-PPG	12,740	1,437	2,748	16,925
WESAD	2,104	297	597	2,998
DALIA	29,294	4,602	5,320	39,216
WildPPG	180,000	15,000	45,000	240,000
AFPPGECG	64,725	43,868	31,843	140,436
Total	793,346	126,205	160,011	1,079,562

B Additional Training Details

Table 11 presents in-domain results across different QA categories. Overall, multimodal LLM-based models consistently outperform the non-LLM baselines across most tasks, demonstrating the effectiveness of integrating pretrained PPG representations with language models for unified physiological inference. In particular, strong performance is observed for tasks such as HR or AF detection, where the underlying physiological patterns are more directly reflected in the PPG waveform. In contrast, lower performance is observed for RR, stress, and SDB detection, suggesting that these tasks remain more challenging due to weaker or more indirect physiological signatures within 10-second PPG segments, higher inter-subject variability, and increased sensitivity to acquisition conditions. Overall, these results highlight both the diversity and challenges of PulseLM, while motivating future research on more robust multimodal physiological understanding.

Table 7: Number of QA samples per category and split.

Category	Train	Val	Test	Total
AF Detection	67,964	44,108	32,560	144,632
Arrhythmia	36,820	4,764	5,243	46,827
Blood Pressure	247,816	30,742	34,549	313,107
HRV-RMSSD	226,828	23,677	41,945	292,450
HRV-SDNN	226,828	23,677	41,945	292,450
HRV-pNN50	226,828	23,677	41,945	292,450
Heart Rate	470,920	51,859	90,459	613,238
Respiratory Rate	31,570	4,181	9,556	45,307
Signal Quality	116,408	14,255	20,838	151,501
Sleep Apnea	205,618	23,525	29,754	258,897
SpO2	17,889	2,563	4,492	24,944
Stress	2,104	297	597	2,998
Total	1,877,593	247,325	353,883	2,478,801

Table 8: Number of QA samples per dataset and split.

Dataset	Train	Val	Test	Total
VitalDB	656,344	82,386	80,711	819,441
UCI	267,162	33,858	34,233	335,253
BCG	1,563	258	192	2,013
PPG-BP	588	72	78	738
SDB	205,618	23,525	29,754	258,897
Sensors	4,893	540	750	6,183
UQVitalSigns	106,031	11,515	36,113	153,659
PPGArrhythmia	36,820	4,764	5,243	46,827
MIMIC PERform	3,239	240	717	4,196
BIDMC	18,824	3,304	2,796	24,924
EarSet	1,368	44	364	1,776
UTSA-PPG	50,960	5,748	10,992	67,700
WESAD	2,104	297	597	2,998
DALIA	29,294	4,602	5,320	39,216
WildPPG	428,060	32,304	114,180	574,544
AFPPGECG	64,725	43,868	31,843	140,436
Total	1,877,593	247,325	353,883	2,478,801

C Discussion

C.1 Unifying PPG Inference

In this work, question answering provides a principled framework for linking raw physiological signals to high-level semantic interpretation. Rather than training separate models for each downstream task, a QA formulation allows heterogeneous physiological attributes to be expressed as questions with interpretable answer choices. This mirrors how clinicians and end users naturally reason about physiological data, for example, by asking whether a heart rate is within a normal range, whether a signal is sufficiently clean for interpretation, or whether a recording indicates health risk.

From a modeling perspective, QA provides several advantages: 1) it enables a unified supervision interface across diverse tasks, eliminating the need for task-specific output heads and loss functions; 2) when answer spaces are constrained to categorical choices, QA reduces ambiguity and supports objective evaluation using standard classification metrics; 3) By associating continuous waveform patterns with discrete semantic concepts, QA also encourages models to learn latent physiological representations that are both interpretable and transferable across tasks.

Table 9: Number of PPG segments by recording environment and split.

Environment	Train	Val	Test	Total
Clinical	503,115	60,957	74,139	638,211
Lab	16,212	1,778	3,709	21,699
In-the-wild	274,019	63,470	82,163	419,652
Total	793,346	126,205	160,011	1,079,562

Table 10: Representative text field examples from PulseLM.

Example	Text Description
1	The patient is a 55-year-old female with a height of 164 cm and a weight of 56 kg, resulting in a body mass index (BMI) of 20.8, which is within the normal range. She has olive skin and engages in physical exercise for approximately 5 hours per week. Currently, her activity level is classified as unknown. The patient’s heart rate is recorded at 67 beats per minute, which falls within the normal heart rate category.
2	The participant is a 27-year-old male with brown (Type V) skin tone. An in-ear PPG signal was recorded from the right ear using a green LED during a running activity. The heart rate was categorized as tachycardia, indicating a heart rate greater than 100 bpm. The recording occurred under full-body movement conditions, which may have introduced expected motion artifacts.
3	The patient presents with tachycardia, with a heart rate exceeding 100 bpm. Blood pressure readings indicate stage 1 hypertension, characterized by a systolic blood pressure within the range of 130-139 mmHg and a diastolic blood pressure between 80-89 mmHg. The quality of the signal is noted as noisy or distorted, which may impact the reliability of the readings. Further evaluation and monitoring may be necessary to assess the patient’s cardiovascular status.
4	The patient exhibits bradycardia with a heart rate recorded at 57 bpm. Blood pressure measurements indicate a normal range. Signal quality is assessed as good. Further evaluation of the cardiac cycle reveals distinct phases, although specific duration measurements are not included in the verified data. Continuous monitoring is recommended to observe any changes in heart rate and overall cardiac function.
5	The patient is a 77-year-old female with a height of 150 cm and a weight of 42 kg, resulting in a BMI of 18.7. Her blood pressure is recorded at 182/80 mmHg, categorizing her in a hypertensive crisis due to the systolic blood pressure exceeding 180 mmHg. Additionally, her heart rate is noted at 101 bpm, indicating tachycardia as it is greater than 100 bpm. There is no significant medical history reported.
6	A 26-year-old male, with a height of 181 cm and weight of 75 kg, resulting in a BMI of 22.9, is currently engaged in meditation. The patient’s emotional state is noted as being associated with meditation, indicating a potential focus on stress reduction or relaxation techniques.

Crucially, a QA-based formulation aligns naturally with multimodal large language models. By conditioning language generation or selection on raw PPG inputs, such models can learn to ground linguistic responses in physiological signals, enabling richer forms of human-machine interaction. However, realizing this potential requires datasets that provide large-scale, physiologically grounded QA supervision across diverse recording conditions.

Motivated by these observations, we design PulseLM to reformulate PPG understanding as a question answering problem. By harmonizing heterogeneous public datasets and converting existing physiological annotations into structured QA supervision, PulseLM aims to provide the missing foundation for language-enabled PPG modeling. In the following section, we describe the construction of this dataset in detail, including data sources, ground truth harmonization, and QA generation procedures.

Table 11: In-domain evaluation across different tasks. Here, the PPG encoder used is PulsePPG (except blind).

Model	HR	BP	SDNN	RMSSD	pNN50	RR	AF	Arrhythmia	SpO ₂	SDB	Stress	SQI	Avg. $\pm$ Std
Blind PPG	83.4	48.3	52.3	46.2	48.9	33.1	85.3	28.9	84.4	24.1	18.3	49.3	50.2 $\pm$ 22.3
Deaf PPG	49.1	30.6	56.9	53.0	55.0	42.4	52.3	42.6	77.9	51.0	1.7	7.8	43.4 $\pm$ 20.3
Fusion PPG	74.9	59.9	73.1	54.4	69.0	39.7	84.8	65.6	72.2	43.0	41.5	85.3	63.6 $\pm$ 15.4
Qwen2.5-7B-Instruct	90.0	60.0	85.3	72.3	78.2	32.7	85.4	59.6	95.4	54.1	49.7	84.7	70.6 $\pm$ 18.4

Instruction-Following Prompt Format used in PULSELM.

[System]

You are a physiological signal analysis expert specializing in PPG-based clinical classification.

Rules:

- Answer MUST be exactly one option from the provided list.
- Output format MUST be strict: `<answer>OPTION</answer>`
- Do not output any extra text.

[User]

Task:

{Question:}

Options:

- *option₁*
- *option₂*
- ...

Return ONLY:

`<answer>OPTION</answer>`

[Assistant]

`<answer>{}/</answer>`

C.2 Limitations and Future Works

Despite its scale and breadth, PulseLM has several points that merit discussion. As presented, PulseLM is constructed from publicly available datasets, which may exhibit variations in population characteristics and label distributions across tasks. To promote broad coverage, we intentionally incorporate datasets containing a range of physiological conditions, including clinically relevant abnormalities, thereby improving representation beyond standard cases. The released dataset and task are naturally challenging, but further gains would also be achieved through more targeted data curation and stronger integration of domain knowledge into model design. It is worth noting that all source datasets used are de-identified originally, with certain metadata that may still carry external information (e.g., age, gender, BMI, and skin tone). Model performance varies across such factors, and these subgroup effects are not investigated in this dataset work.

A further consideration can be the provenance and clinical fidelity of the QA pairs. Because all questions and answers are derived systematically from structured metadata fields in the source datasets rather than from free-form clinical records or expert textual annotations, the generation process is constrained by the label vocabulary and schema of each dataset. This design choice limits hallucination risk and ensures that every answer is traceable to an original dataset attribute, yet does not always capture the full complexity of real-life clinical practice. PulseLM is intended for research purposes on signal-text learning, and models trained on it do not replace clinically validated measurements or expert diagnosis.

At the same time, PulseLM opens several promising directions for future work. One natural extension is the incorporation of additional expert-verified annotations and higher-level clinical summaries to further enhance interpretability and linguistic fidelity. Beyond the current QA setting, PulseLM also includes structured textual information, which provides a foundation for exploring open-ended physiological report generation. In this setting, models could be trained to produce free-form textual summaries grounded in PPG signals and contextual cues, enabling richer and more flexible interaction than discrete QA alone.

Another important direction is multimodal representation learning, where textual supervision from QA pairs or generated reports is leveraged to guide self-supervised or weakly supervised learning of

transferable PPG representations. Such approaches may improve robustness across tasks, datasets, and sensing conditions. Finally, expanding cross-domain generalization benchmarks and developing confidence-aware generation mechanisms remain critical steps toward deploying physiological language models in safety-sensitive health monitoring applications.